

Architectural Particular Specification

UTILITIES CONNECTION CONSTRUCTION AND RECONSTRUCTION OF FOOTWAY AND CARRIAGEWAY

SECTION 1.10

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PART 1 GENERAL

SCOPE OF WORK

1. Where works are located on Tai Yuen Street and Kwok Shui Road, or any other area owned and maintained by the Client, the requirements in this section shall still apply whereas respective government departments and utility companies shall be referred as the Client, where applicable. The Client's requirement regarding the works and associated works of this session shall be deemed to be of the same or a higher standard than the respective government department's standard.

For works located on any other roads, areas, utilities which are not owned, maintained nor under the jurisdiction of the Client, corresponding government departments shall be referred to as stated in this session or otherwise stated.

2. Abbreviations used in this Particular Specification:-

ArchSD	Architectural Services Department
HyD	Highways Department
TD	Transport Department
DSD	Drainage Services Department
WSD	Water Supplies Department
EP	Excavation Permit (The Employer's consent in case of works in areas owned or maintained by the Employer)
CLP	CLP Power Hong Kong Limited

3. The Contractor shall comply with the procedures and standards of the drainage and water supply connections works, cabling works, construction of run-in and reconstruction of damaged footways and carriageway as specified in the Contract and in accordance with statutory requirements and guidelines issued by relevant government departments, authorities and utility companies, and to the satisfaction of the Client.
4. For works involving excavation on carriageway, the Contractor shall employ approved contractors and coordinate with the Sub-contractor for the Works. Contractors on the List of Approved Contractors for Public Works in Group A, B or C for Road & Drainage works are considered to be approved contractors for the drainage connection and run-in/footway work. Similarly, contractors on the same List in Group A, B or C for Waterworks are considered to be approved contractors for the water supply connection work. For water supply connection works not involving any excavation on carriageway, the Contractor shall coordinate with the Sub-Contractor and his Licensed Plumbers to carry out the works.
5. The Contractor shall be responsible for the excavation and formation of any pits and trenches, both temporary or permanent, on the footpath and carriageway for the Sub-Contracts and any other approved Contractors for the construction of lead-in trench, services connection, coordination and diversion of any existing conflicting services as instructed by the Architect at the Contractor's time and cost. The extent and depth of excavation and formation of pits and trenches shall be limited for the identification of the alignment and depth of any underground services installed by the government departments, including HyD, WSD, DSD and all other utility companies, particularly, CLP. The excavation works shall be conducted at the earliest stage of the contract to allow sufficient time for services coordination and connection works. Any delay of works due to late identification of services obstruction shall be borne by the Contractor.

6. The Contractor should bear the cost involved of any administration and application fees as required by the government in connection to the works involved in this section.
7. The Contractor shall apply EP(s) to cover the type(s) of works as necessary or appropriate. Before carrying out any type of the works on Site, the Contractor should conduct utilities circulation and preparation work with relevant government authorities such as TD and Traffic Police, etc. In case resolution of conflicting views between these parties is needed, DSD, HyD and WSD might provide necessary support to the Contractor by the regional chief engineer. Although it might not be a pre-requisite of EP application requiring the Contractor to submit to HyD a coordinated programme for all other involved utility undertakings (UUs) work in the vicinity or to produce documentary evidence that the work proposal has been circulated and the UUs' responses have been received, the Contractor should well plan in advance. Should there be any other utility works being or scheduled to be implemented in the vicinity, the Contractor shall liaise with the relevant UUs so that the works of the Contractor and UUs can be carried out in a coordinated manner. If defects are found during the Maintenance Period after completion of Works, the Contractor shall carry out remedial works unless the Contractor can prove conclusively that the defect was caused by a third party.

PART 2 PRODUCTS

(NOT USED)

PART 3 EXECUTION

3.01 DRAINAGE CONNECTION WORKS

The Contractor shall coordinate with the Sub-Contractor for the submission of connection plans and details to DSD for approval prior to application for an EP. During execution of the works, if amendments to the works outside the building site are required, the Contractor should submit the proposed details to DSD for approval. Minor amendments can be agreed with DSD staff on site.

The Contractor shall submit a Form DSD26 at least 1 month before the technical audit by DSD. Contractor is not required to pay the audit fee in association with the technical audit for the drainage connection works.

The Contractor shall apply for an EP from HyD and shall carry out all necessary preparation works with TD, Traffic Police, etc.

The Contractor shall arrange with DSD for an initial site inspection of the works before backfilling to the pipelines or covering up. One week advance notice by fax to DSD is required for the inspection.

The Contractor shall arrange a handing over inspection with the Architect and DSD when all works for the drainage connections are completed and ready for handing over to DSD. One week advance notice by fax to DSD is required for the inspection.

The Contractor shall submit to DSD as-built record drawings. The drainage works beyond the site boundary and in public roads shall be handed over to DSD for maintenance upon completion.

The Contractor shall provide records of material quality (brand name, grade, class, etc) and acceptance tests (water test, etc) as required by the GS for CE for DSD's scrutiny in the initial site inspection. If an initial inspection cannot be arranged, the Contractor is required to submit a colour closed circuit television survey report for the drainage connection to DSD in order to confirm the internal conditions of the pipelines to be satisfactory.

Design of drainage connection works shall be in accordance with current DSD Sewerage and/or Stormwater Drainage Manual. The pipeline and manholes are to comply with GS for CE and DSD standard drawings.

Unless otherwise specified,

- i. Drainage pipes shall have a minimum diameter of 150mm of the following materials or other materials approved by DSD:-
 1. Foul sewer - vitrified clay, ductile iron, uPVC
 2. Stormwater drain - concrete, ductile iron, Upvc
- B. Foul sewers should be designed so that the velocity of the flow will exceed the self-cleansing velocity on a regular basis. As a general guide, the minimum fall of foul sewers is as follows:-

Diameter of pipe	Minimum fall
150mm	1 in 150
200mm	1 in 200
225mm	1 in 225
250mm	1 in 250
300mm	1 in 300
400mm	1 in 350

C. Manholes should be standard manholes shown in DSD standard drawings.

3.02 RUN-IN CONSTRUCTION AND FOOTWAY RECONSTRUCTION WORKS

1. The Contractor shall write to the respective Regional Offices of HyD stating his intention to carry out the works. He shall also submit an application for EP, stating inter alia, the name of the contractor meeting the requirements in item 1.3 contained herein before, the proposed start and completion date, the name and telephone number of a contact person and attaching a drawing to show the extent and details of the works. The request should be made about 8-10 weeks before the proposed start date.
2. If the application is considered acceptable, HyD will issue an EP within 1 week.
3. The Contractor can then carry out the works on Site.
4. The Contractor shall arrange soil compaction tests and concrete tests as described in clause 3.7 below. The Contractor shall take record photos to meet the HyD's requirements.
5. The Contractor shall provide as-built drawings and a copy of test reports to HyD and arrange a joint handing over inspection with the Architect and HyD. He shall give at least 2 working days advance notice by fax to HyD for the inspection. He shall carry out any remedial works as identified by HyD during the inspection.
6. HyD will take over the completed works if in order.
7. The Contractor shall arrange the following tests to be carried out by an independent laboratory accredited under the Hong Kong Laboratory Accreditation Scheme. He shall submit the test reports to HyD before the handing over inspection.
 - A. Relative compaction of the sub-grade material underneath footways and run-in according to clause 6.68 of GS for CE.
 - B. Relative compaction of sub-base beneath run-in according to clause 9.44 of GS for CE.
 - C. Compressive strength of concrete according to clause 16.58 to 16.62 of GS for CE.
 - D. Bituminous material coring tests according to clause 9.58 to 9.61 of GS for CE at the rate of 1 core for every 500m² of carriageway area or part thereof.

3.03 WATER SUPPLY CONNECTION WORKS

1. The Contractor shall consult WSD for the location of supply connection when enquiring on water availability.
2. The Contractor shall submit to WSD for approval an application letter for carrying out the works outside the building site enclosed with the following:-
 - A. a plan showing the alignment of proposed connection main;
 - B. the inside/fire services works proposal;
 - C. the name of the Approved Contractor employed for the execution of the Works; and
 - D. completed form WWO 542.

This submission should be given to WSD at least 3 months before the date scheduled for making tee connection described in below sub-clause g. It is advisable to submit the proposal at an early stage well in advance for WSD to check its compliance with the Waterworks requirements. Re-submission(s) may be required as only proposals meeting the Waterworks requirements will be approved.
3. The Contractor shall settle the demand note to be issued by WSD and send a copy of the receipt to WSD.
4. WSD might issue an approval letter and a demand note within 21 working days for payment of connection charge.

- 5 The Contractor shall apply to HyD for an EP for execution of supply connection works and to WSD for the tee connection works.
- 6 The Contractor shall excavate up to the proposed tee connection point to expose the existing supply main and notify WSD, with the Architect's endorsement, by fax at least 7 working days in advance to inspect the exposed main.
 - A. Lay and extend the connection main back into the building site. (The pipe should be terminated at an appropriate location and capped at the end to prepare for 'pick-up' connection under below sub-clause 4.11).
 - B. Notify WSD to inspect jointly with the Architect the completed pipeline before backfilling.
 - C. Backfill the completed pipeline (the Contractor shall provide compaction test results for WSD's checking).
 - D. In the presence of the Architect or the Architect's Representative and WSD staff:-
 - i. carry out pressure test, in accordance with Clause 23.77 of the GS for CE after the fill material has been deposited and compacted over the complete length of the pipeline;
 - ii. clean and sterilise the connection main; and pick up the connection with the tee valve outside the building site.
 - E. Carry out permanent road reinstatement, including the area around the connection tee, to the satisfaction of HyD.
 - F. Notify WSD to inspect jointly with the Architect the connection tee valve pit. The Contractor may choose to invite WSD for checking the pit during the final inspection under below sub-clause 4.11 if he has confidence in its workmanship. He should ensure proper adjustment of the level of the cover and that no debris is accumulated inside the pit. Any improper pit construction may hinder the turning on of water supply.
7. The Contractor shall submit to WSD as-built record drawing showing the alignment and level of connection main within 2 weeks of completion of pipeworks.
8. The Contractor shall submit WSD Part IV of Form WWO 46 to request for inspection upon completion of all inside/fire services works.
9. After all inside/fire services works are checked to be satisfactory by WSD, the Contractor shall, in the presence of the Architect or the Architect's Representative and WSD staff:-
 - i. flush to clean the connection main; and
 - ii. pick up the outstanding final connection within the building site.
10. The Contractor shall hand over the connection main works beyond the building site to WSD upon completion.
11. The Contractor shall give prior notice to WSD by fax at least 2 working days before the date ready for joint inspection under above sub-clause 4.8b, d & f and above sub-clause 4.11.
12. The Contractor shall comply with Section 23 of the GS for CE and the following specification in executing the water supply connection works.
13. Identification tapes shall be laid continuously along the connection mains with the printed pattern facing upwards. The tapes shall be as shown on Drawing No. WSD 1.31.
14. Detectable warning tapes shall be laid continuously along the buried non-metallic connection mains and be used in conjunction with the identification tapes. The tape shall be placed directly on top of the non-metallic mains with the printed pattern facing upwards and fixed onto the pipe body as shown on Drawing No. WSD1.37. The tape shall be brought to the surface within the connection tee valve pit and mounted on the pit wall at a terminal point as per details shown on the Drawing

15. Unless otherwise specified,

- A. Backfill to pipelines shall be suitable material with a moisture content $\pm 3\%$ of the optimum moisture content and free from clay lumps and stones retained on a 20 mm sieve.
- B. Fill material within 300 mm above the pipe shall be deposited in 100 mm layers and compacted by hand rammers to obtain a relative compaction of at least 95% throughout.
- C. Fill material on top of the 300 mm layer above the pipe shall be deposited and compacted in layers to obtain a relative compaction of at least:-
 - i. within 150 mm of road formation level - 98%
 - ii. elsewhere - 95%
- D. Thrust and anchor blocks shall be designed to take the full thrust and constructed in Grade 20/20 concrete.
- E. The whole section of new fresh water connection main shall be sterilised by filling with water that has been dosed with chloride of lime at a minimum concentration of 30 ppm.
- F. After at least 30 minutes, the sterilized water in the pipe shall be drained off and the pipe be flushed with potable water to the satisfaction of WSD staff before picking up the final connection.

3.04 WORK IN ROADS

1 Approval for temporary traffic arrangements and control

- A. In addition to the requirements stated in the Contract, temporary traffic arrangements shall be in accordance with conditions and restrictions imposed by the Commissioner for Transport and the Commissioner of Police, and temporary lighting, signing, guarding and traffic control arrangements shall be in accordance with the requirements and restrictions in the Code of Practice for the Lighting, Signing and Guarding of Road Works.
- B. The Contractor shall make all arrangements with and obtain the necessary approvals from the Commissioner for Transport, the Commissioner of Police, the Director of Highways and any other relevant authority for temporary traffic arrangements and control.
- C. The Contractor shall submit detailed working proposal to the Commissioner of Police, Commissioner for Transport, the Architect and any other relevant authorities for approval at least 4 weeks prior to the intended commencement date of any temporary traffic arrangements. Corresponding Traffic Impact Assessment on works on sensitive routes including traffic delay and queue length shall also be submitted when required. Where major traffic diversions or changes to no-stopping restrictions which need to be gazetted or changes to signal equipment setting and siting are required, the time between submission of the proposals and implementation will be increased accordingly. The Contractor's attention is drawn to the time required for compiling the Traffic Impact Assessment (depends on the complexity of the works) and the arrangement of gazette notice (may take more than 12 months). The Contractor shall consider the time requirement in his programme of works.
- D. No work shall be permitted to commence until necessary agreements and approvals have been given.

2 Temporary traffic arrangements and controls

- A. Temporary traffic diversions and pedestrian routes shall be provided where work in roads or footways obstruct existing vehicular or pedestrian access. The relevant work shall not commence until the approved temporary traffic arrangements and control have been implemented.
- B. Temporary traffic arrangements and control for work in roads and footways shall comply with the requirements contained in the current edition of the document

'Code of Practice for Lighting, Signing and Guarding of Road Works' issued by the Hong Kong Government; a copy of the document shall be kept on the Site.

- C. Temporary traffic light signals shall be a type approved by the Commissioner for Transport and shall comply with the requirements contained in the current editions of the documents 'Type Approval Procedure for Portable Traffic Light Signals' and 'Specification for Vehicle Actuated/Fixed Time Portable Traffic Signal Equipment' issued by the Hong Kong Government.
- D. Temporary traffic signs, including posts, backing plates and faces, shall comply with the requirements for traffic signs contained in Section 12 of GS for CE except as stated in the following paragraphs v. & vi.
- E. The thickness of backing plates for temporary traffic signs which will be erected for less than 6 months may be reduced to 1.5mm; the posts for the signs may be constructed of timber or other material provided that in the opinion of the Architect the traffic signs will be stable and safe.
- F. The arrangement of information contained on sign faces for temporary traffic directional signs shall be designed by the Contractor and to be approved by relevant authorities. The details of the background, borders and legends, including letters, numerals, characters and symbols, shall comply with the requirements of the Commissioner for Transport.
- G. Temporary traffic arrangements and control shall be inspected and maintained regularly, both by day and night. Traffic lights, lights and signs shall be kept clean and easy to read. Equipment which is damaged, dirty, incorrectly positioned or not in working order shall be repaired or replaced immediately.
- H. The Contractor shall ensure that movements of the construction traffic within the Site are properly supervised and controlled, and where required by the Architect the Contractor shall, after seeking approval from the relevant authorities, provide manually operated traffic control at all points where construction traffic enters or crosses a public road.
- I. Electric lamps complying with BS 3143 shall always be used when connection to an electricity main is readily available and the voltage shall be stepped down to not more than 50 volts. When an electricity supply is not available battery lamps shall be used. Kerosene lamps shall only be used with the approval of the Architect as a last resort and in emergencies.

3 Particulars of temporary traffic arrangements and control

The following particulars of the proposed temporary traffic arrangements and control shall be submitted to the Architect for information at least 7 days before the traffic arrangements and control are implemented:-

- A. Details of traffic diversions and pedestrian routes;
- B. Details of lighting, signing, guarding and traffic control arrangements and equipment, and
- C. Any conditions or restrictions imposed by the Commissioner for Transport, the Commissioner of Police, the Director of Highways or any other relevant authority, including copies of applications, correspondence and approvals.

4 Use of roads and footways

- A. Roads and footways on the Site in which work is not being carried out shall be maintained in a clean and passable condition and shall not be used to store materials or park Constructional Plant or other vehicles.
- B. Measures shall be taken to prevent excavated material, silt or debris from entering drainage systems in roads and footways; entry of water to the gullies shall not be obstructed.
- C. Surfaced roads on the Site and leading to the Site shall not be used by tracked vehicles unless protection against damage is provided.

- D. Constructional Plant and other vehicles leaving the Site shall be loaded in such a manner that excavated material, mud or debris will not be deposited on roads; roads shall be covered or protected to prevent dust being emitted. The wheels of Constructional Plant and other vehicles shall be washed when necessary before leaving the Site to avoid the deposition of mud and debris on roads.
- 5 Work in roads and footway (within site)
- A. Work in roads on the Site shall be carried out in sections such that the length of road occupied at any time does not exceed that stated in the Contract and the width of road occupied at any time does not exceed the width of one traffic lane unless permitted by the Architect. Work in each section shall be completed and the road shall be reinstated and opened to traffic before work commences in the next section. Work in any section, including loading and unloading, shall be carried out in such a manner that traffic and utilities in the adjacent road and pedestrian access in the adjacent footway will be adequately maintained.
- B. Before excavations are carried out in roads or footways, except in areas covered with paving blocks or tiles, the limits of the area to be reinstated shall be bounded by a continuous saw-cut groove. The groove shall be at least 6 mm wide and at least 50 mm deep. Cutting the groove and breaking up the road or footway shall be carried out in such a manner that the adjacent road or footway, including edges, are not damaged.
- C. Excavated material shall not be stored adjacent to excavations in roads or footways unless permitted by the Architect.
- D. Vehicular access across excavations in roads shall be provided with steel covers. The covers shall be designed to BS 449: Part 2 and shall be capable of withstanding the full traffic load permitted to use the road. The covers shall be secured in position and shall have anti-skid coating with skid resistance values measured in accordance with BS 3262 of not less than 45. Sufficient steel covers shall be kept on the Site adjacent to excavations in roads to permit vehicular access across the excavations in case of emergency.
- 6 The following additional conditions shall be observed during work execution in existing carriageways:-
- A. The Contractor shall complete the works in all existing carriageways in an efficient and expeditious manner and shall hand back the occupied carriageways to the Employer at the earliest opportunity.
- B. The Contractor shall not disturb any roadside trees/shrubs and shall take precautions against damage to roadside amenity planting in his main laying work. Should it be necessary to remove any roadside trees for main laying, the Contractor shall give at least six weeks advance notice to the Architect for the arrangement of transplanting of the roadside trees by the Employer.
- C. A minimum width of 1m shall be maintained for pedestrian movements on partially closed footways.
- D. The Contractor's attention is drawn to the requirement that the extent of works in public highways shall be limited to a maximum of 100 m length for each Excavation Permit. The length includes the length of permanent works and any necessary working space.
- 7 Reinstatement of roads and footways
- A. Temporary diversion, pedestrian access and lighting, signing, guarding and traffic control equipment shall be removed immediately when they are no longer required. Roads, footways and other items affected by temporary traffic arrangements and control shall be reinstated to the same condition before the work started or to such other condition as may be agreed or instructed by the Architect.

END OF SECTION